

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

$M \cup N$ $\wedge \vee$ $\langle X, S \rangle / \langle e^{2\pi i}, e^{2\pi i} \rangle$
 π_0, π_1^2 klein.

$\mathbb{Z} \times \mathbb{Z}$ $\text{klein. free product.}$
 $\langle a, b \mid aba = b \rangle$ $ababab \dots$
 $aba^2b^2 = e$

$ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$ $\langle a, b \mid ab = ba \rangle$
 $\mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^2$ $T: \mathbb{Z} \times \mathbb{Z}$
 $T = \mathbb{Z} \times \mathbb{Z}$ $\mathbb{Z} \times \mathbb{Z}$

Theorem $S^1 \vee S^1$
 $M: n\text{-dim}$ $\mathbb{R}^3: 0$

$\pi_1(M) \cong \pi_1(M \cup \dots)$
 $\pi_1(RP^n) \cong \mathbb{Z}_2$ $\pi_1(S^n) = 0$

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